

REVERSER SHIFTS ON GENERALIZED DIHEDRAL GROUPS: CUBIC ZETA COMPRESSION AND TWO-PERIOD RIGIDITY

ANONYMOUS

ABSTRACT. Let A be a finite abelian group and $G = \text{Dih}(A) = A \rtimes \{\pm 1\}$. We form a one-step shift on alphabet G by declaring $g \rightarrow h$ when $hgh^{-1} = g^{-1}$. Write $N = |A|$, $t = |A[2]|$, and $c = N/t$. We prove that the shift is mixing and that its entire $2N$ -state adjacency spectrum depends only on (N, t) . When $N > t$, the characteristic polynomial is

$$\lambda^{2N-c-2}(\lambda - t)^{c-1}(\lambda^3 - 2t\lambda^2 + t(t - N)\lambda - Nt(N - t)).$$

Thus its zeta function is controlled by one repeated linear factor and a cubic. In particular,

$$\#\text{Fix}(\sigma) = N + t, \quad \#\text{Fix}(\sigma^2) = t(3N + t).$$

These first two counts recover (N, t) . Conversely, an explicit cosetwise graph model shows that equal parameter pairs give isomorphic relation graphs. Hence two shifts in the family are conjugate exactly when their pairs $(|A|, |A[2]|)$ agree. Nonisomorphic groups, for example $\mathbb{Z}/9\mathbb{Z}$ and $(\mathbb{Z}/3\mathbb{Z})^2$, therefore produce the same symbolic system. Exact group-law enumeration checks the quotient, spectrum, periods, and collapse on the registered finite families.

1. INTRODUCTION

An element h reverses g when conjugation by h sends g to g^{-1} . Reversing symmetries are standard in group-theoretic approaches to dynamics; their structural setting is developed, for example, by Baake and Roberts [1]. Generalized dihedral groups are a basic source of such involutory behavior. Their commuting graphs have also been studied directly [3].

This note asks a different symbolic question. Instead of forming an undirected commuting graph, we use the directed local relation $hgh^{-1} = g^{-1}$ as the adjacency rule of a one-dimensional shift of finite type. A small-group calculation gives two immediate anomalies: the adjacency matrix has very low rank, and nonisomorphic abelian groups with the same order and the same size of their two-torsion subgroup give identical graphs.

We close these observations for every finite abelian A . The contributions are:

- (1) a canonical row-and-coset model depending only on $(N, t) = (|A|, |A[2]|)$;
- (2) mixing and unique maximal entropy measure for every member;
- (3) a complete characteristic polynomial, periodic-point formula, and rational zeta function compressed to a cubic factor;
- (4) family rigidity from only the first two periodic counts.

The concepts of reversing elements, generalized dihedral groups, equitable spectral compression, and finite-type zeta functions are prior tools [1, 2, 4]. The residual object is the particular directed reverser relation and its exact symbolic package. A bounded search through 28 August 2026 found no exact-system collision under the relation and parameter keywords recorded in the package audit. That is not an exhaustive priority search, and no absolute priority claim is made.

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2. THE RELATION SHIFT

Write A additively and let

$$G = \text{Dih}(A) = A \rtimes \{0, 1\}, \quad (a, \epsilon)(b, \eta) = (a + (-1)^\epsilon b, \epsilon + \eta \bmod 2).$$

Elements $(a, 0)$ are rotations and $(a, 1)$ are reflections. Put

$$T = A[2] = \{a \in A : 2a = 0\}, \quad N = |A|, \quad t = |T|, \quad c = N/t.$$

Since T is a subgroup, t divides N and c is a positive integer.

Definition 2.1 (reverser shift). Let M_A be the zero-one matrix indexed by G with

$$M_A(g, h) = 1 \iff hgh^{-1} = g^{-1}.$$

The reverser shift is

$$X_A = \{x \in G^{\mathbb{Z}} : M_A(x_i, x_{i+1}) = 1 \text{ for every } i\},$$

with left shift σ .

The local relation is directed: the role of the state being reversed and the role of its reverser are not interchangeable in the definition.

Lemma 2.2 (successor classes). *The outgoing neighbors have the following form.*

- (1) If $g = (a, 0)$ with $a \in T$, every element of G follows g .
- (2) If $g = (a, 0)$ with $a \notin T$, precisely the N reflections follow g .
- (3) If $g = (a, 1)$, its successors are the t rotations $(y, 0)$ with $y \in T$ and the t reflections $(y, 1)$ with $y \in a + T$.

Proof. Conjugation by a rotation fixes every rotation, while conjugation by a reflection inverts every rotation. This gives the first two statements. For a reflection, direct multiplication gives

$$(y, 0)(a, 1)(y, 0)^{-1} = (a + 2y, 1),$$

and

$$(y, 1)(a, 1)(y, 1)^{-1} = (2y - a, 1).$$

Every reflection is an involution, so equality with $(a, 1)^{-1} = (a, 1)$ requires $2y = 0$ in the first display and $2(y - a) = 0$ in the second. $\square$

Proposition 2.3 (mixing). *Every X_A is a mixing shift of finite type. It therefore has a unique measure of maximal entropy.*

Proof. Every reflection has the identity rotation among its successors. Every non-two-torsion rotation reaches a reflection and then the identity, while a two-torsion rotation already lies in that class. Conversely, the identity rotation has every group element as a successor. The graph is strongly connected. The identity rotation also has a loop, so the graph period is one and its adjacency matrix is primitive. The maximal-measure conclusion is the standard Parry theorem [5]. $\square$

3. CANONICAL COLLAPSE AND SPECTRUM

Partition the vertices into the t two-torsion rotations R_T , the $N - t$ remaining rotations R_U , and the reflections. Further partition the reflections into the c cosets of T in A . By lemma 2.2, the graph is canonically described as follows:

- every vertex of R_T points to all $2N$ vertices;
- every vertex of R_U points to all reflections;
- every reflection in coset C points to all of R_T and all reflections in C .

In particular, the graph depends only on (N, t) .

We use the row action on column functions, $(M_A f)(g) = \sum_h M_A(g, h)f(h)$. On functions constant on the three coarse classes, the adjacency action is represented by

$$(1) \quad Q_{N,t} = \begin{pmatrix} t & N-t & N \\ 0 & 0 & N \\ t & 0 & t \end{pmatrix}.$$

Theorem 3.1 (complete spectral compression). *If $N > t$, then*

$$(2) \quad \det(\lambda I - M_A) = \lambda^{2N-c-2}(\lambda - t)^{c-1}(\lambda^3 - 2t\lambda^2 + t(t - N)\lambda - Nt(N - t)).$$

In particular, $\text{rank } M_A = c + 2$. If $N = t$, then M_A is the $2N$ by $2N$ all-ones matrix, so its characteristic polynomial is $\lambda^{2N-1}(\lambda - 2N)$.

Proof. Assume $N > t$. Vectors whose coordinate sum is zero within R_T , within R_U , or within any single reflection coset are killed by M_A . Their direct sum, denoted V_0 , has dimension

$$(t - 1) + (N - t - 1) + c(t - 1) = 2N - c - 2.$$

Next take the space V_t of functions that vanish on all rotations and are constant with value α_i on the i th reflection coset, where $\sum_i \alpha_i = 0$. lemma 2.2 shows that M_A multiplies this $(c - 1)$ -dimensional space by t . Finally, let V_Q be the three-dimensional space of functions constant on R_T , on R_U , and on the union of all reflections. These three invariant spaces are mutually disjoint and their dimensions sum to

$$(2N - c - 2) + (c - 1) + 3 = 2N.$$

Thus $\mathbb{C}^G = V_0 \oplus V_t \oplus V_Q$, and the action on V_Q is $Q_{N,t}$. Direct expansion gives

$$\det(\lambda I - Q_{N,t}) = \lambda^3 - 2t\lambda^2 + t(t - N)\lambda - Nt(N - t).$$

The cubic has nonzero constant term because $N > t$, so $Q_{N,t}$ has rank three. Since $t > 0$, the action on V_t is invertible. The zero action on V_0 therefore gives rank $(c - 1) + 3 = c + 2$, as well as the factorization. If $N = t$, every element of A has order at most two, G is abelian of exponent two, and every pair satisfies the relation; hence M_A is all ones. $\square$

For $N > t$, the entropy is $\log \rho_{N,t}$, where $\rho_{N,t}$ is the Perron root of the cubic factor in (2). The full-shift endpoint has entropy $\log(2N)$.

4. PERIODIC POINTS, ZETA, AND TWO-COUNT RIGIDITY

Let $F_k(A) = \# \text{Fix}(\sigma^k) = \text{tr}(M_A^k)$. The spectral decomposition gives, for $N > t$,

$$(3) \quad F_k(A) = (c - 1)t^k + \text{tr}(Q_{N,t}^k).$$

Theorem 4.1 (zeta and first periods). *For $N > t$, the Artin–Mazur zeta function of the shift is*

$$(4) \quad \zeta_{X_A}(z) = \frac{1}{(1 - tz)^{c-1} [1 - 2tz + t(t - N)z^2 - Nt(N - t)z^3]}.$$

At the full-shift endpoint $N = t$,

$$(5) \quad \zeta_{X_A}(z) = \frac{1}{1 - 2Nz}.$$

Moreover, for every N, t , including $N = t$,

$$(6) \quad F_1 = N + t, \quad F_2 = t(3N + t).$$

Proof. For an SFT, $\zeta(z) = \det(I - zM_A)^{-1}$ [4]. Substitution of (2), or direct evaluation of the determinant on the invariant decomposition, gives (4). If $N = t$, the all-ones matrix has one eigenvalue $2N$ and all others zero, proving (5).

For $N > t$, $\text{tr } Q_{N,t} = 2t$ and $\text{tr}(Q_{N,t}^2) = 2t(N+t)$. Adding the $(c-1)$ copies of the eigenvalue t gives

$$F_1 = (c-1)t + 2t = N + t, \quad F_2 = (c-1)t^2 + 2t(N+t) = t(3N+t).$$

The same formulas reduce to $(2N)$ and $(2N)^2$ when $N = t$. $\square$

Theorem 4.2 (family conjugacy rigidity). *For finite abelian groups A and B ,*

$$X_A \cong X_B \iff (|A|, |A[2]|) = (|B|, |B[2]|).$$

The reverse implication is realized by a one-block conjugacy induced by a relation-graph isomorphism.

Proof. A conjugacy preserves every periodic count. Put $S = F_1 = N + t$. By (6), t solves

$$(7) \quad 2x^2 - 3Sx + F_2 = 0.$$

The genuine root is a positive integer and satisfies $t \leq N$, hence $t \leq S/2$. The other root is $3S/2 - t \geq S$, so it cannot satisfy the same bound. Thus t is recovered uniquely and $N = S - t$. Conjugate shifts therefore have the same (N, t) .

Conversely, the canonical description preceding (1) uses only the sizes t , $N - t$, and c reflection fibers of size t . Choose bijections between the two R_T classes and the two R_U classes, choose a bijection between their sets of reflection cosets, and then choose a bijection inside each paired coset. The three adjacency bullets show entry by entry that the result is a directed graph isomorphism, hence a one-block conjugacy. $\square$

For example, $A = \mathbb{Z}/9\mathbb{Z}$ and $B = (\mathbb{Z}/3\mathbb{Z})^2$ are nonisomorphic, but both have $(N, t) = (9, 1)$, so their reverser shifts are conjugate. The symbolic system forgets the abelian group structure beyond order and two-torsion size.

5. EXACT CONTROLS AND SCOPE

The accompanying Python program independently implements the semidirect product, inverse, conjugation, and relation graph for 20 product-of-cyclic presentations. It compares every canonical adjacency entry, explicitly checks the zero, t , and quotient invariant spaces, checks rank and strong connectivity, verifies traces through period ten, computes small characteristic and zeta polynomials, reconstructs (N, t) from (F_1, F_2) , and tests nonisomorphic same-parameter collapses both at $(N, t) = (9, 1)$ and $(16, 4)$.

These finite computations guard the group-law convention and all endpoint formulas; they are not evidence for claims beyond the proofs. The classification is only inside this reverser-shift family. It does not classify generalized dihedral groups, reversing symmetry groups, commuting graphs, or arbitrary group-relation SFTs. External release and priority claims remain on hold.

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